

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

Dec 2023

Course Code: MEC101

Course: BASIC MECHANICAL ENGINEERING

QP No. :UD041-1

Duration: 3 hours

Max. Marks:100

PART – A

Answer all the questions

10 x 2 = 20 Marks

1. A conical enlargement produced using a drilling machine accommodates the head of a screw. Mention the name of the operation to make a conical enlargement and give details.
2. The shaper's ram moves back and forth horizontally. Is this machine tool capable of producing an angular surface? If yes, write about the operation. If no, state the reason.
3. In one of the forming processes, the force is applied to the billet using the movement of the ram inside the container. Identify the process and draw the schematic diagram.
4. Cleaning is the last step in the casting process to create a final product. List a few operations performed in the cleaning process to get a final product.
5. Distinguish between AC welding and DC welding.
6. State any four essential properties of ideal refrigerants.

7. Mention the fundamental difference between a two-stroke and four-stroke engine.
8. Lubrication is the process of applying the lubricant between the surfaces of contact of two moving parts. Point out the functions of the lubricating system.
9. Compare the non-renewable energy resources with renewable energy resources.
10. List out the types of steam turbines.

PART – B

Answer all the questions

4 x 15 = 60 Marks

11. List all the operations that can be performed on a lathe. A single-point cutting tool is used for some of these operations. In some other operations, a multi-point cutting tool is used. With the help of neat sketches, explain two operations that can be performed using a single-point cutting tool and two operations that can be performed using a multi-point cutting tool.
- (OR)**
12. The various hand-forging tools are used to perform hand-forging operations. Describe any five hand-forging operations with suitable sketches and mention the hand-forging tools used in each operation.
13. A pattern is a replica of the object to be made by the casting process. A single-piece pattern is one of the patterns used for making simple-shaped casting. Explain the step-by-step procedure of the green sand moulding process to prepare the mould cavity using a single-piece pattern with suitable sketches.

(OR)

14. In one of the welding processes, the required heat to melt the metal parts is supplied by a flame. Identify the name of the welding process. List the equipment needed to perform that welding process and give the details about the types of flames with neat sketches.
15. One of your friends has purchased a window air conditioner. He wants to know the working of a window air conditioner. To fulfill his curiosity, explain the working principle of the window air conditioner with a suitable sketch.

(OR)

16. The fuel supply system supplies fuel into the cylinder at the correct proportion for combustion. In a diesel engine, a fuel injection pump and fuel injector are used to inject the correct quantity of fuel into the cylinder. With relevant sketches, describe the working of the fuel injection pump and fuel injector.
17. Draw a schematic layout of a gas turbine power plant and explain the function of integrated parts of the plant. Also, state the advantages of the gas turbine power plant.

(OR)

18. With a neat sketch, explain the working of the Benson boiler in detail. Also, mention the advantages of the Benson boiler compared with the La-Mont boiler.

PART – C

Answer the following

1 x 20 = 20 Marks

19. A cooling system is necessary for the internal combustion engine to prevent the overheating of its parts and extend the engine's life and for the steam power plant to reduce the temperature of the steam. With the aid of neat sketches, explain the following cooling systems:

- a) Air cooling system used in I.C engine. (6)
- b) Water cooling system used in I.C. engines. (7)
- c) Cooling water circuit used in the steam power plan. (7)
